

1. Administrative & Study Identification

Full Study Title: A Phase 1, Open-label, Randomized, Four-Period Crossover Study to Evaluate the Relative Bioavailability of Evobrutinib Tablet Manufacturing Batches in Healthy Participants under Fasted Conditions **Short Title/Acronym:** Relative Bioavailability of Evobrutinib Tablet Batches **Protocol Number:** MS200527_0131 **NCT Number:** NCT07214922 **Sponsor Name:** Merck KGaA (EMD Serono) / Merck Healthcare KGaA **Investigational Product:** Evobrutinib (M2951) **Phase of Development:** Phase 1 **Medical Monitor:** Merck KGaA Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the Pharmacokinetics (PK), safety, and tolerability of different manufacturing batches of the M2951 (evobrutinib) tablet formulation relative to a reference batch under fasted conditions in healthy participants. **Secondary Objectives:** To evaluate the overall safety and tolerability profiles across the different drug batches.

2.2 Background & Rationale To address and understand the variations in the production of evobrutinib tablets. This study generates essential pharmacological data to verify that variations in manufacturing batches do not significantly impact the drug's relative bioavailability, systemic exposure, or safety profile compared to the reference batch.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Reference Batch - 4-period crossover design) **Blinding:** Open-label **Randomization:** Randomized (Sequences: ABCD, BDAC, CADB, DCBA)

3.2 Planned Enrollment & Duration Target \$N\$: Not explicitly provided (Standard Phase 1 healthy volunteer cohort) **Number of Sites:** Single/regional Phase 1 Clinical Unit **Estimated Study Start:** 18-Jan-2023 **Estimated Primary Completion:** 24-Feb-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Overtly healthy adults (males and females) with no clinically significant abnormalities identified

on physical examination or laboratory evaluation. 2. Body weight within 50.0 and 100.0 kg (inclusive) and BMI within 19.0 to 30.0 kg/m2 (inclusive). 3. Stable non-smokers for at least 3 months preceding Screening. Female participants agree to use appropriate contraception and barrier methods.

4.2 Exclusion Criteria 1. Clinically relevant findings in biochemistry, hematology, or urinalysis (e.g., ALT/AST above ULN, Creatinine above normal limits). 2. Positive tests for Hepatitis B, Hepatitis C, HIV I/II, or SARS-CoV-2 at screening. 3. History of clinically relevant respiratory, gastrointestinal, renal, hepatic (including hemochromatosis, Wilson's disease), cardiovascular, psychiatric, or chronic liver disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single oral dose of Evobrutinib (Treatment A, B, C, or D) administered per period under fasted conditions. **Route of Administration:** Oral tablet. **Duration of Treatment:** 4 treatment periods (single doses on Days 1, 3, 5, and 7) with a 48-hour washout period between each treatment period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard Phase 1 unit dietary provisions (meals controlled and standardized around the fasted dosing periods). **Prohibited:** Alcohol consumption from 48 hours prior to first administration; live vaccines or live-attenuated virus vaccines within 3 months prior to Day 1.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Medical History, Vitals, COVID-19/Viral testing, Blood/Urine Labs
Baseline/Dosing	Days 1, 3, 5, 7	Randomized sequential single-dose administration of Batches A, B, C, D (48h washout in between), PK Blood Sampling
Interim	N/A	Intensive Phase 1 Confinement Monitoring
End of Study	Follow-up	Final safety labs, Vital Signs, Exit Evaluation, AE monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Pharmacokinetics (PK) of Evobrutinib, specifically measuring Area Under the Concentration-time Curve (AUC) and Maximum Observed Plasma Concentration (Cmax) to evaluate relative bioavailability. **Measurement Method:** Plasma concentration analysis via serial blood sampling.

7.2 Secondary & Exploratory Endpoints Incidence and severity of Adverse Events (AEs) to measure safety and tolerability. Changes

from baseline in vital signs, ECGs, and clinical laboratory parameters.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during Phase 1 confinement, including ECGs and clinical laboratory testing. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours to the sponsor). **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by Merck KGaA.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: PK Analysis Set for bioavailability comparison; Safety Set for all participants receiving at least one dose. **Sample Size Justification:** Powered to establish bioequivalence (evaluating if the 90% confidence intervals for geometric mean ratios of PK parameters fall within standard equivalence margins). **Handling of Missing Data:** Standard Phase 1 PK protocols (e.g., plasma concentrations below the lower limit of quantification [LLOQ] treated as zero prior to C_{max}).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous onsite monitoring and strict adherence to protocol within the Phase 1 clinical facility. **Audit Trail:** eCRF locking, tracking of strict PK sample chain-of-custody, and source data verification (SDV).

11. Study Identifiers & Governance

Primary ID: MS200527_0131 **Registry Number:** NCT07214922 **Sponsor/Lead Organization:** Merck KGaA (EMD Serono) **Study Title:** Relative Bioavailability of Evobrutinib Tablet Batches **Official Regulatory Title:** A Phase 1, Open-label, Randomized, Four-Period Crossover Study to Evaluate the Relative Bioavailability of Evobrutinib Tablet Manufacturing Batches in Healthy Participants under Fasted Conditions

12. Clinical Framework (Study Specific)

Primary Condition: Healthy Volunteers **Diagnosis Threshold:** Overtly healthy status (BMI 19.0-30.0 kg/m², Body Weight 50-100 kg) **Medical Methodology:** Pharmacokinetic (PK) Crossover

Evaluation **Optimization Target:** Cmax and AUC bioequivalence among manufacturing batches

13. Study Procedures & Methodology

Management Pathway: Phase 1 clinical pharmacology unit confinement with fasted dosing **Education Protocol (HCP):** Phase 1 protocol training and rigorous dosing sequence (ABCD, BDAC, CADB, DCBA) management **Education Protocol (Patient):** Informed consent, unit confinement rules, and dietary restriction guidelines **Quality Audit Mechanism:** PK sample reconciliation and source data verification

14. Observation & Follow-Up Schedule

Total Duration: Approx. 4 weeks (Screening + 7 days of treatment periods + safety follow-up) **Onsite Visit Cadence:** Confinement from Day -1 through the completion of Day 7 PK sampling **Remote Monitoring Frequency:** N/A (In-patient unit) **Checklist Review Interval:** Daily during the confinement and active PK sampling periods

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				